ALEEM NAEEM

Machine Learning & Embedded Systems Engineer Lahore, Pakistan | ☎ +92 315 6733883 | ✉ aleemnaeem53@gmail.com

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PROFESSIONAL SUMMARY

Innovative **Electrical and Computer Engineering** specialist with a focus on merging **Machine Learning** with **Intelligent Embedded Systems**. Proven track record in developing ML pipelines for remote sensing and optimizing robotic control through sensor data processing. Expert in Python and C++ with a deep interest in real-time edge computing.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), C++, SystemVerilog.
- **Machine Learning:** SVM, Random Forest, XGBoost, Clustering, PCA, Feature Engineering.
- **Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch.
- **Embedded & Robotics:** Arduino, Raspberry Pi, RTOS, Sensor Fusion, IoT Edge Computing.
- **Deployment & Tools:** Flask, FastAPI, Docker, Git, Jupyter Notebooks.

EXPERIENCE

Embedded Software Engineer (Freelance) | *Remote* * Architected real-time, multithreaded embedded applications, improving system responsiveness for client projects.

- Streamlined hardware-software integration by debugging low-level firmware in C++.

Robotics & IT Instructor | *Lahore, Pakistan* * Mentored students in Robotics, Arduino, and Python programming.

- Designed hands-on curriculum focused on building autonomous systems and pattern recognition logic.

KEY PROJECTS

Ship Classification Using Remote Sensing Data * Developed an end-to-end ML pipeline utilizing **PCA** for dimensionality reduction and **SVM** for classification.

- Achieved high-performance benchmarks across Accuracy, Precision, and F1-score for satellite imagery analysis.

Robotics & Intelligent Embedded Systems * Implemented sensor-based control systems using ML algorithms to optimize robot navigation and pattern recognition.

- Integrated real-time data processing to reduce latency in robotic decision-making.

Smart Energy Monitoring System (IoT) * Built an IoT-based monitoring solution featuring **edge computing** to process energy consumption data locally before cloud transmission.

EDUCATION

MS Electrical & Computer Engineering (In Progress) University of Engineering & Technology (UET), Lahore

BS Electrical Engineering University of Management and Technology (UMT), Lahore

- *Achievement: Full Merit-Based Academic Scholarship*

CERTIFICATIONS & AWARDS

- **SystemVerilog Accelerated Verification using UVM** – Cadence
- **Chief Minister's Merit-Based Laptop Award** (Academic Excellence)